

17 An object of mass m has a constant acceleration due to a constant resultant force F .

In a distance s , the speed of the object is increased from u to v .

What is the relationship between s , m , F , u and v ?

A $Fs = mv^2 - mu^2$

B $2Fs = mv^2 - mu^2$

C $v^2u = u^2 - \left(\frac{F}{m}\right)s$

D $v^2 = u^2 - 2\left(\frac{F}{m}\right)s$